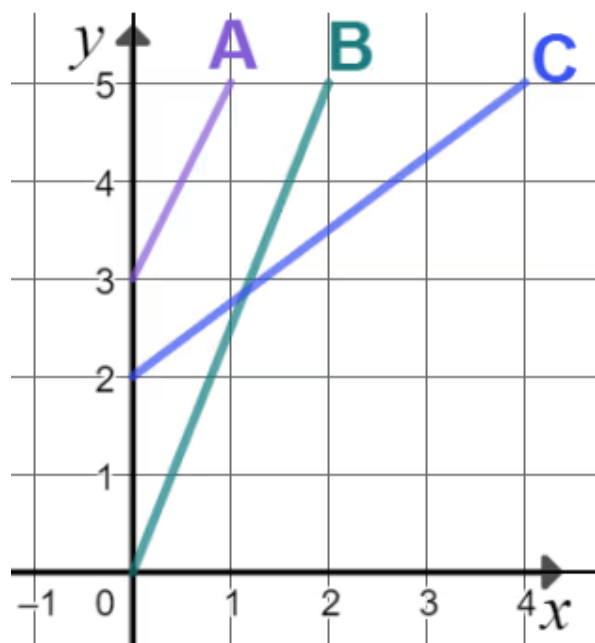




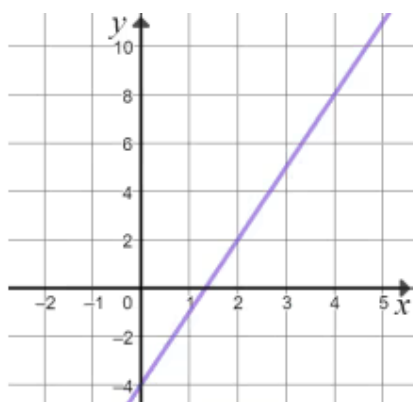
Modelling with graphs

- 1 The rate of change is how quickly one variable changes in relation to another. Fill in the blank
- 2 Which of these lines has the steepest gradient? Tick 1 correct answer



- ☒ Line a
☐ Line b
☐ Line c

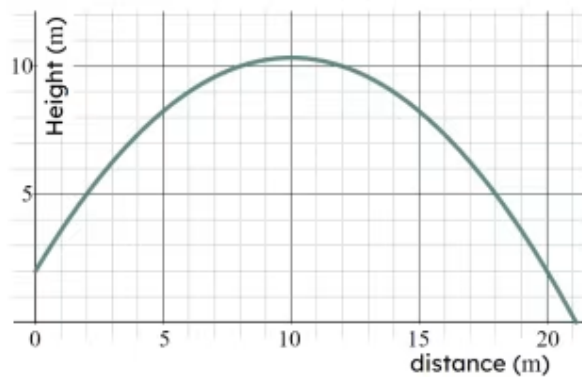
- 3 Look at this graph. What is the value of x when $y = 5$? Tick 1 correct answer



- ☐ $x = 1$
☐ $x = 2$
☒ $x = 3$
☐ $x = 4$



- 4 This parabola models the flight of a cricket ball being thrown. Estimate the maximum height of the cricket ball. Tick 1 correct answer



- ☐ 9.8 metres
☐ 10 metres
☒ 10.2 metres
☐ 11 metres
☐ 21 metres

- 5 Laura is plotting the graph of the line $3y + 4x = 12$. She only wants to plot two points. Which of these points would be the best points for Laura to use? Tick 2 correct answers

- ☒ When $x = 0$
☐ When $x = 1$
☐ When $x = 2$
☒ When $x = 3$
☐ When $x = 4$

- 6 Jun wants to draw the graph of $y = 0.4x + 3.2$ for positive values of x . The coordinates of the first point on the line that has positive integer coordinates are (2, 4). Fill in the blank

